

APEXPLANET

CYBERSECURITY & ETHICAL HACKING INTERNSHIP PROGRAM

TASK – 2

VULNERABILITY ASSESSMENT REPORT

Nessus Essentials

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Course / Program	Cybersecurity & Ethical Hacking Internship Program
Authorized Target	Metasploitable 2 – 192.168.56.104
Assessment Tool	Tenable Nessus Essentials

Prepared from practical Nessus scan evidence captured in the authorized laboratory environment.

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1. Introduction

This report documents the vulnerability assessment performed as part of Task 2 – Network Security & Scanning of the ApexPlanet Cybersecurity & Ethical Hacking Internship Program. The assessment was carried out against the authorized Metasploitable 2 virtual machine using Tenable Nessus Essentials.

The purpose of the assessment was to identify vulnerabilities and security weaknesses exposed by the target, review their severity, examine the technical details reported by Nessus, and record appropriate remediation guidance. The assessment was restricted to the controlled Host-Only laboratory environment.

2. Assessment Objective

To perform a vulnerability assessment of the authorized Metasploitable 2 target using Nessus Essentials, identify reported security findings, classify them by severity, and document their security impact and remediation.

3. Laboratory Environment and Scope

Item	Details
Attacker / Analyst VM	Kali Linux
Authorized Target	Metasploitable 2
Target IP	192.168.56.104
Vulnerability Scanner	Tenable Nessus Essentials
Assessment Scope	Authorized laboratory target only

All active vulnerability-scanning activity documented in this report was directed at the Metasploitable 2 laboratory system. No public or third-party systems were used as assessment targets.

4. Nessus Setup and Scan Configuration

Nessus was installed and initialized on Kali Linux. The setup workflow included selecting Nessus Essentials, creating the local administrator account, allowing Nessus to initialize and download/compile its plugins, and opening the Nessus web interface.



Welcome to Nessus

You can click Settings to configure the Nessus proxy, plugin feed, and encryption password settings before you start the installation, or you can select Register Offline to configure an offline installation.

When you are ready, click Continue to proceed with the installation.

☐ Register Offline

Settings

Continue

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Figure 1. Nessus welcome screen during initial setup.



Welcome to Nessus

Choose how you want to deploy Nessus. Select an option to get started.

- ☒ Set up a purchased instance of Nessus
- ☐ Start a trial of Nessus Expert
- ☐ Start a trial of Nessus Professional
- ☐ Register for Nessus Essentials
- ☐ Set up Nessus Essentials Plus for Education
- ☐ Link Nessus to another Tenable product

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Continue

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Figure 2. Nessus edition selection, including Nessus Essentials.



Create a user account

Create a Nessus administrator user account. Use this username and password to log in to Nessus.

Username *

Password *

[Back](#)[Submit](#)

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Figure 3. Nessus local administrator account setup.

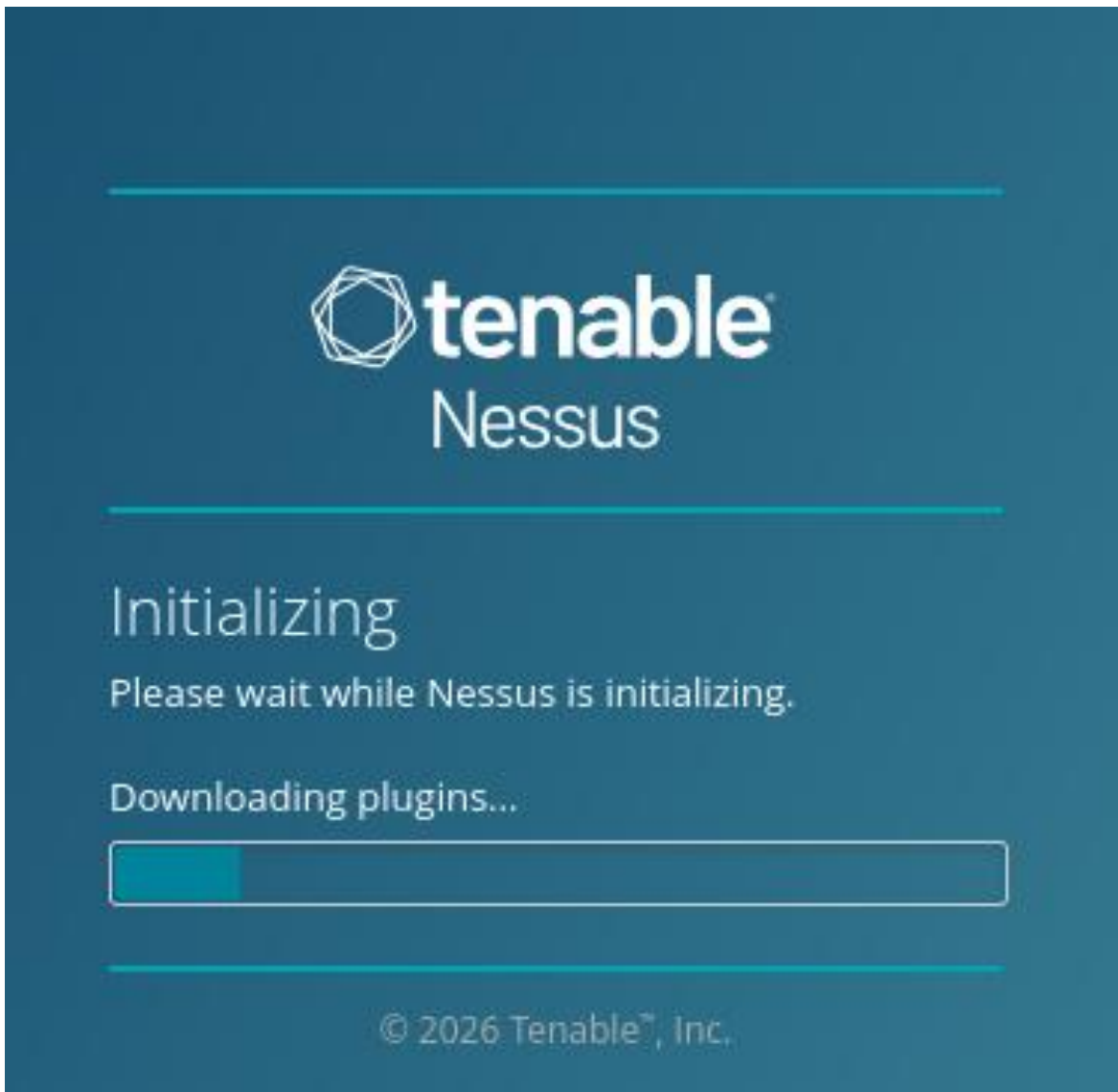


Figure 4. Nessus initialization and plugin download.

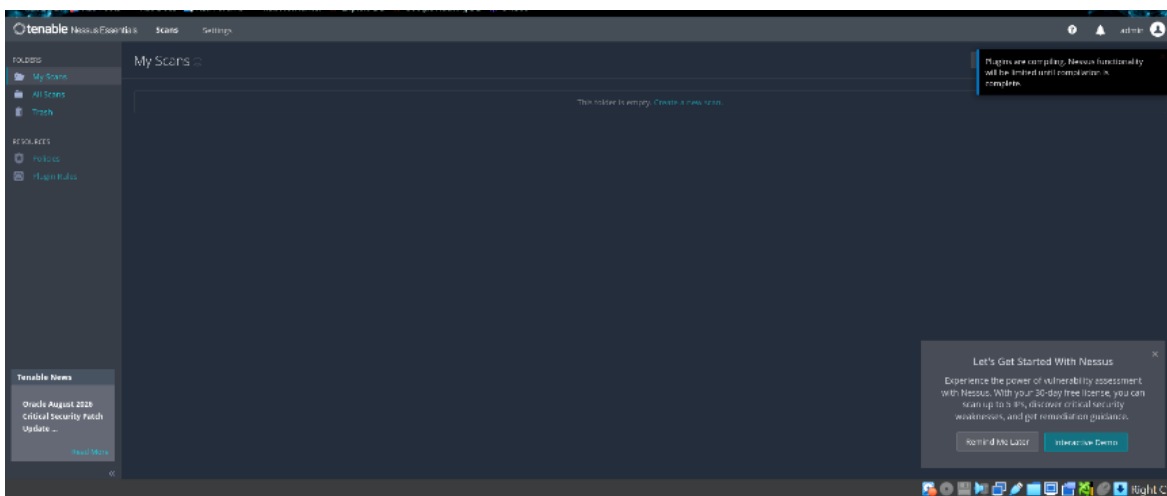


Figure 5. Nessus web interface after initialization.

A Nessus vulnerability scan was configured for the authorized target 192.168.56.104. The scan was named “Metasploitable 2 Vulnerability Scan.” The completed assessment produced findings across multiple severity levels.

5. Vulnerability Assessment Results

The Nessus results identified vulnerabilities in the target environment at High, Medium, and Low severity, along with informational findings. The detailed findings below are taken from the captured Nessus results.

Severity	Finding	Plugin ID	CVSS	Affected Service / Port
High	Samba Badlock Vulnerability	90509	7.5	445/tcp (CIFS)
Medium	ISC BIND Denial of Service	136808	5.9	DNS
Low	ICMP Timestamp Request Remote Date Disclosure	10114	2.1	ICMP

5.1 High – Samba Badlock Vulnerability

Nessus reported a High-severity Samba Badlock vulnerability (Plugin ID 90509) affecting the CIFS/SMB service on port 445/tcp. The captured Nessus details describe an improper authentication-level negotiation condition that can allow a man-in-the-middle attacker to force a downgrade and execute arbitrary Samba network calls.

Security impact: according to the Nessus finding, successful exploitation could allow an attacker to view or modify sensitive Active Directory data or disable services, depending on the affected environment.

Reported solution: upgrade Samba to a patched release. Nessus specifically lists fixed releases including 4.2.11, 4.3.8, and 4.4.2 or later.

Evidence: Nessus reported that the required patch was not applied on the assessed host.

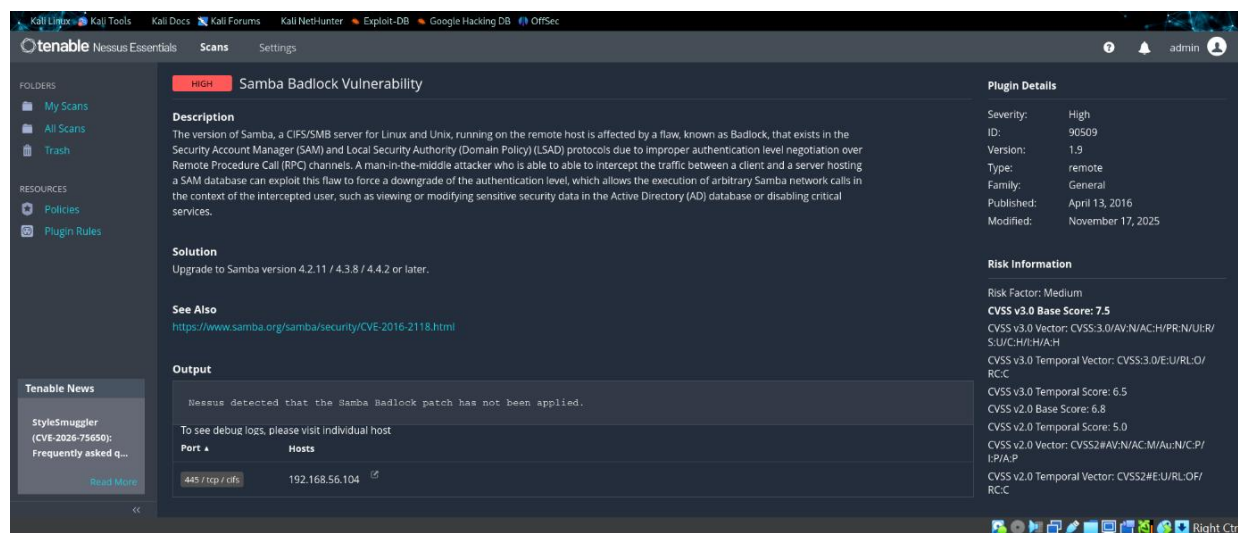


Figure 6. Nessus detailed finding for the Samba Badlock vulnerability (High).

5.2 Medium – ISC BIND Denial of Service

Nessus reported a Medium-severity ISC BIND Denial of Service finding (Plugin ID 136808) in the DNS family. The finding has a CVSS v3.0 base score of 5.9. Nessus reported that versions including BIND 9.4.2 are affected and described a condition in which an unauthenticated remote attacker may cause the DNS service to stop responding.

The Nessus output indicated that the installed version was 9.4.2 and listed 9.11.19 as a fixed version for the reported issue. The scanner noted that it relied on the application-reported version rather than testing exploitation.

Reported solution: upgrade BIND to an appropriate patched release.

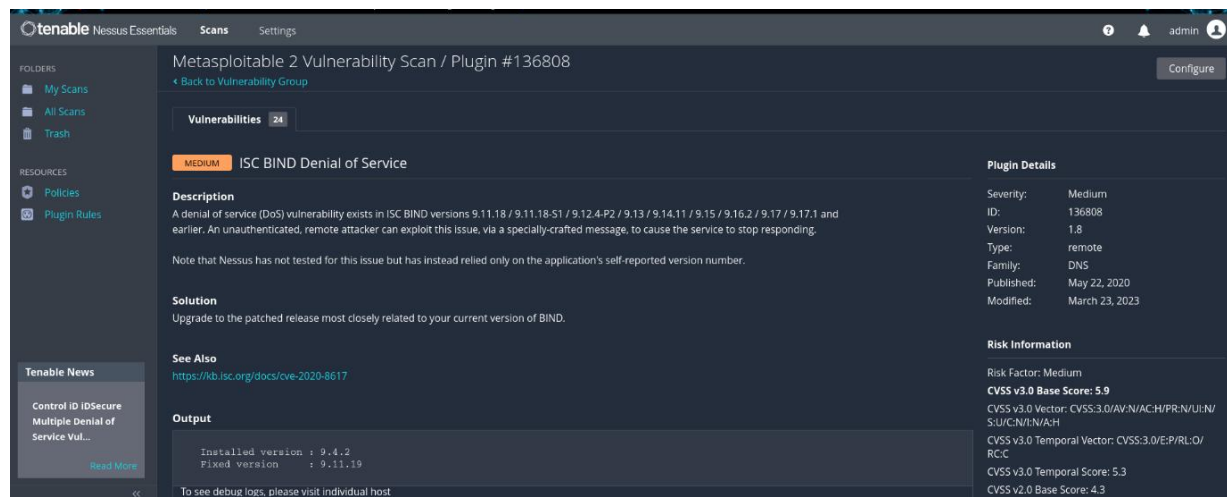


Figure 7. Nessus detailed finding for ISC BIND Denial of Service (Medium).

5.3 Low – ICMP Timestamp Request Remote Date Disclosure

Nessus reported a Low-severity ICMP Timestamp Request Remote Date Disclosure finding (Plugin ID 10114) with a CVSS v2.0 base score of 2.1. The finding is associated with CVE-1999-0524 and indicates that ICMP timestamp requests/replies can disclose the target system's local time.

Reported solution: filter ICMP timestamp requests and replies where this functionality is not required.

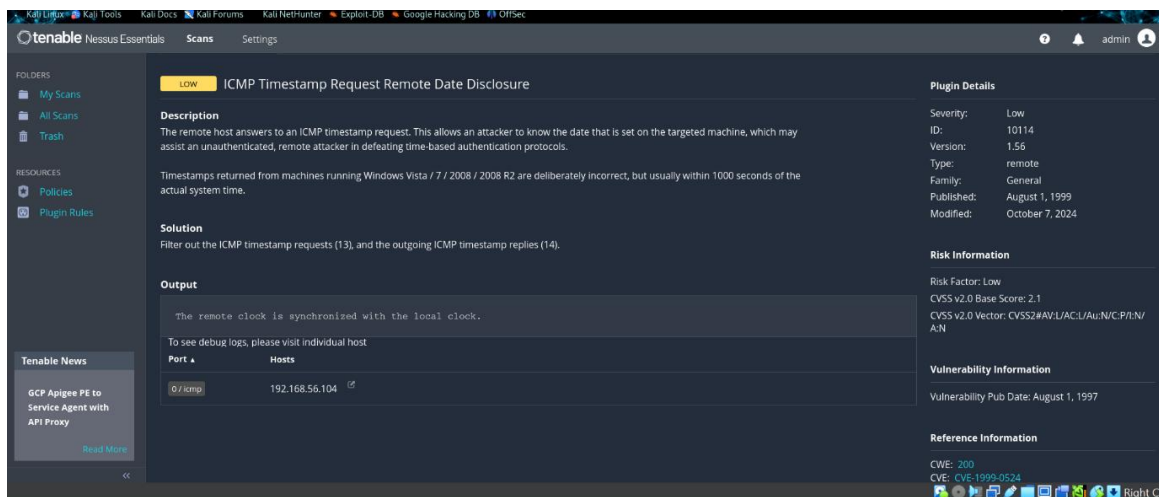


Figure 8. Nessus detailed finding for ICMP Timestamp Request Remote Date Disclosure (Low).

6. Risk Analysis and Recommended Remediation

The assessment shows that the Metasploitable 2 system exposes multiple outdated or vulnerable services. Because Metasploitable is intentionally vulnerable, these results are expected in a training environment; however, they demonstrate the types of weaknesses that should be remediated on production systems.

Severity	Finding	Risk Interpretation	Recommended Action
High	Samba Badlock	Potential compromise or manipulation of sensitive Samba/AD-related operations through authentication downgrade conditions.	Upgrade Samba to a patched supported release and reduce unnecessary SMB exposure.
Medium	ISC BIND DoS	An unauthenticated remote condition may cause the DNS service to stop responding.	Upgrade BIND to a patched release and restrict unnecessary DNS exposure.
Low	ICMP Timestamp Disclosure	May disclose system time information useful for reconnaissance.	Filter ICMP timestamp requests/replies when not required.

Priority should be given to the High-severity Samba finding, followed by the Medium-severity BIND issue. The Low-severity ICMP issue can be addressed through network filtering and hardening.

7. Conclusion

The Nessus vulnerability assessment of the authorized Metasploitable 2 system was successfully completed. The scan identified security findings across High, Medium, and Low severity levels, demonstrating the risks associated with outdated network services and insecure configurations.

The most significant captured finding was the High-severity Samba Badlock vulnerability, followed by the Medium-severity ISC BIND Denial of Service issue and the Low-severity ICMP timestamp disclosure. The results were reviewed from the Nessus plugin details and documented with their reported remediation guidance.

Appendix A – Nessus Evidence

The following evidence was captured during the Nessus assessment. The screenshots are retained as practical proof of setup, scan execution, vulnerability findings, and the final results view.

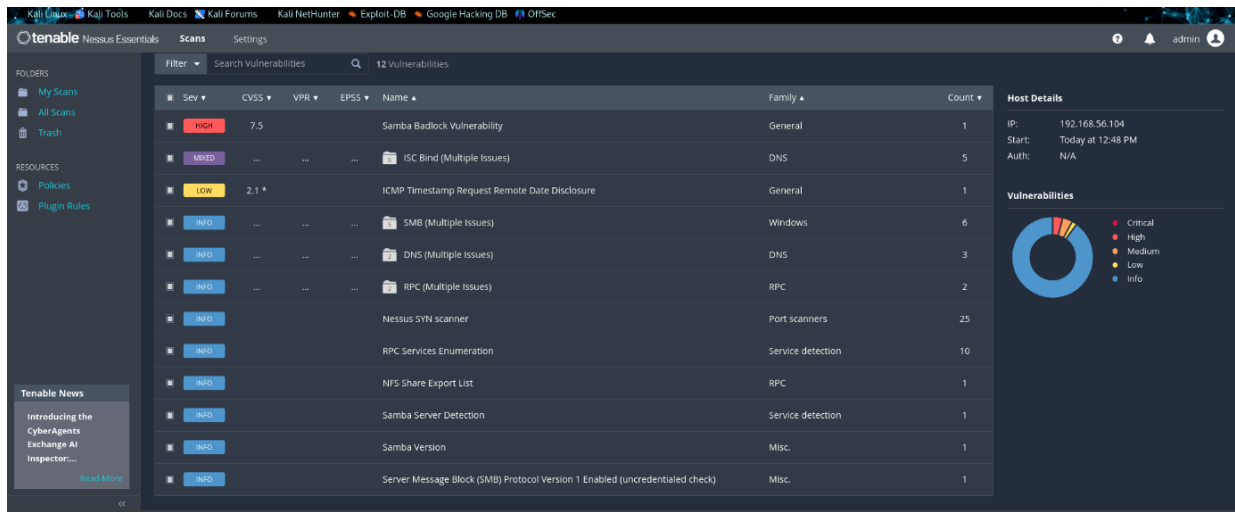


Figure 9. Nessus assessment evidence.

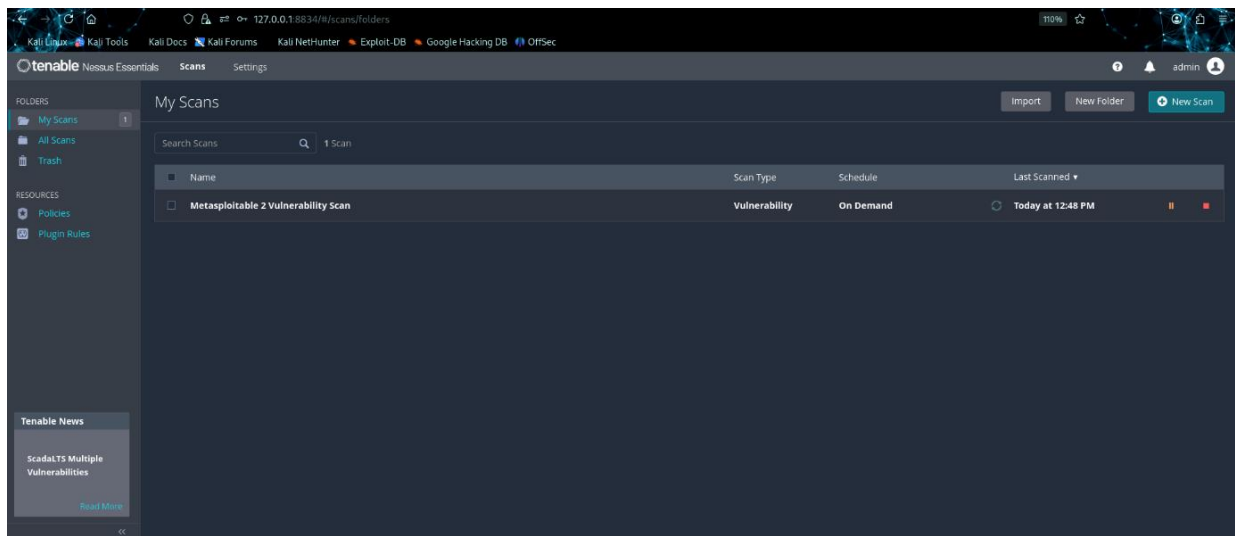


Figure 10. Nessus assessment evidence.